1. Given $f(x) = (7x + 3)e^x + \ln(x + 4)$. Find $f'(x)$, $f''(x)$.

2. Given the total revenue $R(x) = 52x - 0.5x^2$ and the total cost $C(x) = 30x + 1$.

a) Find the marginal revenue function.

b) Find the maximum profit and the number of units that must be produces and sold in order to yield this maximum profit.

3. A company is designing an open-top, rectangular container that will be twice as long as it is wide and must have a volume of 6 m^3. The cost of making the bottom of container is 5$ per square meter, and the cost of the sides is 7$ per square meter. Find the dimensions of the container that will minimize total cost.

4. Evaluate the integral $\int (3x - 2)e^x dx$.

5. A firm's marginal revenue and marginal cost functions are given by

$$MR = R'(x) = 240 - 0.6x^2 \quad \text{and} \quad MC = C'(x) = 150 + 0.3x^2.$$

Fixed costs are 50 and $R(0) = 0$.

a) Write down an expression for total revenue $R(x)$.

b) Write down an expression for the total cost function $C(x)$.

c) Determine the maximum profit.

6. Given the demand function $D(x) = \frac{120}{\sqrt{x+32}}$ and the supply function $S(x) = 10\sqrt{x}$. Calculate

a) The equilibrium point. **b)** The consumer surplus at the equilibrium point.

c) The producer surplus at the equilibrium point.

7. Find the relative maximum and minimum values of

$$f(x, y) = 2x^3 - 6xy - 12x + 3y^2 + 1.$$

8. Consider the following data showing the average life expectancy of men in various years.

Year	1980	1990	2000	2010	2020
Life expectancy of men, (years)	67.6	70.1	73.8	75.2	76.1

a) Find the regression line.

b) Use the regression line to predict the life expectancy of men in 2030.